

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

TIDE INTERNATIONAL (USA), INC.,
Petitioner,

v.

UPL NA INC.,
Patent Owner.

IPR2020-01113
Patent 7,473,685 B2

Before SUSAN L. C. MITCHELL, ZHENYU YANG, and
MICHAEL A. VALEK, *Administrative Patent Judges*.

VALEK, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining Some Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

Tide International (USA), Inc., (“Petitioner”) filed a Petition (Paper 2, “Pet.”), seeking *inter partes* review of claims 1–4 and 7–12 of U.S. Patent No. 7,473,685 B2 (Ex. 1001, “the ’685 patent”). We instituted trial on all of the grounds in the Petition. Paper 12 (“Institution Dec.”).

Following institution, UPL NA, Inc., (“Patent Owner”) filed a Response (Paper 14, “Resp.”), Petitioner filed a Reply (Paper 21, “Reply”), and Patent Owner filed a Sur-Reply (Paper 23, “Sur-Reply”). We held a hearing on October 27, 2021, and a transcript is of record. Paper 29 (“Tr.”).

We have jurisdiction pursuant to 35 U.S.C. § 6. This Decision is a Final Written Decision under 35 U.S.C. § 318(a) and 37 C.F.R. § 42.73. Petitioner bears the burden of proving unpatentability of the challenged claims by a preponderance of the evidence. *See* 35 U.S.C. § 316(e); 37 C.F.R. § 42.1(d) (2019).

Upon consideration of the full record, and for the reasons explained below, we find Petitioner has shown by a preponderance of the evidence that claims 1–4 are unpatentable, but has not shown by a preponderance of the evidence that claims 7–12 are unpatentable on the grounds in the Petition.

II. BACKGROUND

A. *Real Parties in Interest*

Petitioner identifies itself, “Tide International (USA), Inc.,” as well as “Zhejiang Tide Cropsience Co. Ltd., Ningbo Tide Imp. & Exp. Co., Ltd., and Tide International Company Limited” as the real parties-in-interest. Pet. 73. Petitioner explains that Tide International Company Limited is its parent corporation and that “the remaining parties are defendants in

co-pending litigation.” *Id.* Patent Owner identifies itself, “UPL NA Inc.,” as the real party-in-interest. Paper 3, 1.

B. Related Matter

Petitioner and Patent Owner identify the following related matter involving the ’685 patent: *UPL NA Inc. v. Tide International (USA), Inc.*, No: 8:19-cv-1201-RSWL-KS (C.D. Cal. 2019) (“the related litigation”).

C. The ’685 Patent

The ’685 patent was issued on January 6, 2009, and claims priority to a utility application filed on April 22, 2002, and a provisional application filed December 18, 2001. Ex. 1001, codes (45), (63), (60).

The ’685 patent relates to “[d]ry flow, low compact, dust free, soluble granules” containing the pesticide “acephate” in combination with other ingredients, and processes for making such. Ex. 1001, Abstr. According to the Specification, “[t]he formulation of acephate presently in use . . . poses the problems of dust, low pourability, high transportation costs, high capital manufacturing investment, measurement difficulties, difficulties in packing material disposal, handling problems, high risk of caking and others.” *Id.* at 2:13–20.

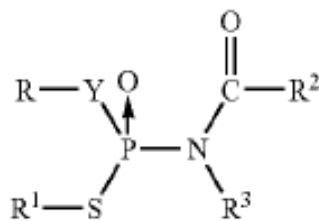
The Specification purports to overcome these problems by describing a method that constitutes “an improvement over prior manufacturing processes” that produces granules that are “more advantageous than prior granular products and exhibit[] certain very desirable characteristics.” Ex. 1001, 2:33–43.

The claims of the '685 patent recite particular formulations of granule compositions “consisting of” acephate and certain recited ingredients in amounts falling within specified ranges. Ex. 1001, 7:44–8:65.

D. Challenged Claims

The Petition challenges claims 1–4 and 7–12. Of these, claims 1 and 7 are independent. Claim 1 reads as follows:

1. A chemically stable dry flow, low compact, dust free soluble phosphoramidothioate granule consisting of
- (i) 85-98% w/w an insecticidally active compound of the following formula:



wherein R and R¹ individually are alkyl, alkynyl or alkenyl group containing up to 6 carbon atoms, R² is hydrogen, an alkyl group containing 1 to 18 carbon atoms, a cycloalkyl group containing 3 to 8 carbon atoms, an alkenyl group containing 2 to 18 carbon atoms or an alkynyl group containing 3 to 18 carbon atoms, R³ is hydrogen or an alkyl group containing 1 to 6 carbon atoms, and Y is oxygen or sulfur, wherein said insecticidal active compound is Acephate:

- (ii) 0.1-5.0% w/w a dispersing agent;
(iii) 0.1-3% w/w a wetting agent;
(iv) 0.01-0.08% w/w an antifoaming agent;
(v) 0.01-1% w/w a stabilizer and
(vi) fillers to make 100%,
wherein said granule has a length of 1.5-3.0 mm and a diameter of 0.5-1.5 mm.

Ex. 1001, 7:44–8:4. Claim 7 is directed to a granule with the same dimensions, ingredients, and ranges as claim 1 and that additionally consists

of a “binding agent” and a “disintegrating agent” within specified ranges.
See id. at 8:19–48.

E. Instituted Grounds of Unpatentability

Petitioner asserts the following grounds of unpatentability:

Claims Challenged	35 U.S.C. §	References
1–4, 7–12	103(a) ¹	Misselbrook, ² CN ’588, ³ JP ’902, ⁴
1–4, 7–12	103(a)	Misselbrook, Mayer, ⁵ CN ’588
1–4, 7–12	103(a)	Misselbrook, JP ’902, Mayer

Petitioner relies on the declaration of William Geigle (Ex. 1003) in support of these grounds. Patent Owner relies on the declaration of David A. Rockstraw, Ph.D., P.E. (Ex. 2007), which was submitted with its Response.

III. ANALYSIS OF ASSERTED GROUNDS

A. Legal Standards

“In an [*inter partes* review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is

¹ The Leahy-Smith America Invents Act, Pub. L. No. 112-29, 125 Stat. 284 (2011) (“AIA”), included revisions to 35 U.S.C. §§ 102 and 103 that became effective after the filing of the application that led to the ’685 Patent. Therefore, we apply the pre-AIA versions of 35 U.S.C. §§ 102 and 103.

² U.S. Patent No. 6,387,388 B1, issued May 14, 2002 (Ex. 1005) (“Misselbrook”).

³ CN 1127588A, published July 31, 1996 (Ex. 1006). Petitioner has provided a certified translation (Ex. 1007), which we refer to and cite herein as “CN ’588.”

⁴ JP 9-315902, published Dec. 9, 1997 (Ex. 1008). Petitioner has provided a certified translation (Ex. 1009), which we refer to and cite herein as “JP ’902.”

⁵ U.S. Patent No. 6,030,924, issued Feb. 29, 2000 (Ex. 1010) (“Mayer”).

unpatentable.” *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016). This burden of persuasion never shifts to patent owner. *See Dynamic Drinkware, LLC v. Nat’l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015).

A claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the claimed subject matter and the prior art are such that the subject matter, as a whole, would have been obvious at the time of the invention to a person having ordinary skill in the art. *KSR Int’l Co. v. Teleflex, Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and (4) objective evidence of nonobviousness, if any.⁶ *See Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

Subsumed within the *Graham* factors is the requirement that the skilled artisan would have had a reasonable expectation of success in combining the prior art references to achieve the claimed invention. *Pfizer, Inc. v. Apotex, Inc.*, 480 F.3d 1348, 1361 (Fed. Cir. 2007). “Obviousness does not require absolute predictability of success . . . all that is required is a reasonable expectation of success.” *In re O’Farrell*, 853 F.2d 894, 903–4 (Fed. Cir. 1988). Moreover, “[t]he combination of familiar elements according to known methods is likely to be obvious when it does no more than yield predictable results.” *KSR*, 550 U.S. at 416. “If a person of

⁶ In this case, Patent Owner does not direct us to any objective indicia of nonobviousness.

ordinary skill can implement a predictable variation, § 103 likely bars its patentability.” *Id.* at 417.

On the other hand, a patent claim “is not proved obvious merely by demonstrating that each of its elements was, independently, known in the prior art.” *KSR*, 550 U.S. at 418. An obviousness determination requires finding “both ‘that a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.’” *Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1367–68 (Fed. Cir. 2016) (citation omitted); *see KSR*, 550 U.S. at 418 (for an obviousness analysis, “it can be important to identify a reason that would have prompted a person of ordinary skill in the relevant field to combine the elements in the way the claimed new invention does”). A petitioner’s assertion of obviousness “cannot employ mere conclusory statements. The petitioner must instead articulate specific reasoning, based on evidence of record, to support the legal conclusion of obviousness.” *In re Magnum Oil Tools Int’l, Ltd.*, 829 F.3d 1364, 1380 (Fed. Cir. 2016) (citing *KSR*, 550 U.S. at 418).

B. Level of Ordinary Skill in the Art

We begin by assessing the level of skill in the art. Considerations relevant to determining the level of ordinary skill in the art include the type of problems encountered in the art and prior art solutions to those problems. *See Envtl. Designs, Ltd. v. Union Oil Co.*, 713 F.2d 693, 696–97 (Fed. Cir. 1983). The prior art itself may also reflect an appropriate level of skill. *Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001).

According to Petitioner’s declarant, Mr. Geigle, “a person of ordinary skill in the art for the ’685 patent has a bachelor’s degree or Ph.D. in chemistry or chemical engineering, with at least two to four years of experience or education specifically in the formulation and development of [solid pesticides subjected to granulation processes].” Ex. 1003 ¶ 18. Mr. Geigle additionally opines “a person who does not satisfy the identified educational level may still qualify as a person of ordinary skill if they [have] had more relevant work experience.” *Id.*

Patent Owner’s declarant, Dr. Rockstraw, offers a similar description of a person of ordinary skill in the art (“POSITA”), but states the POSITA’s experience more generally as “experience with agrochemicals and related formulations.” Ex. 2007 ¶ 77. According to Dr. Rockstraw,

requiring at least two to four years of experience or education “specifically in the formulation and development” of “solid pesticides subjected to granulation processes” is overly limiting. Ex. 1003, ¶ 18. There are other areas of practice in which an understanding of particles, granules, dispersion, wetting, foaming, and/or stability are important, and these fields overlap considerably.

Id. ¶ 79.

We adopt Patent Owner’s description of a POSITA because it reflects the experience of one familiar with the types of problems and solutions described in the ’685 patent and the cited prior art, which include granule as well as other kinds of agrochemical formulations. That said, we do not discern a meaningful difference between the parties’ proposals. Our findings and conclusions would be the same under Petitioner’s description of a POSITA.

C. Claim Construction

In an *inter partes* review, claim terms are “construed using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. [§] 282(b), including construing the claim in accordance with the ordinary and customary meaning of such claim as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent.” 37 C.F.R. § 42.100(b).

Here, the district court in the related litigation construed certain terms that appear in the challenged claims. Ex. 1015. The district court’s constructions are noted in the table below.

Claim term	District court’s construction
“dispersing agent”	an agent that assists with dispersion
“antifoaming agent”	an agent that reduces or prevents the formation of foam
“stabilizer”	an agent that promotes physical and chemical stability

Id. at 19. At Patent Owner’s urging, the district court also determined that the preamble of claim 1 was not limiting. *Id.* at 10. Petitioner adopts these constructions, and further contends that the identical preamble in claim 7 is not limiting, for purposes of this proceeding. Pet. 3.

In addition to the terms construed by the district court, Petitioner contends the following terms should be given their “plain and ordinary meaning” as set forth in the table below and evidenced by the testimony of Petitioner’s expert.

Claim term	Petitioner’s proposed construction
“wetting agent”	an agent that when added to a liquid, reduces the interfacial tension between the liquid and the surface on which it is spreading

“binding agent”	an agent that assists in the binding of particles together in a formulation
“disintegrating agent”	an agent that enables a liquid to penetrate the pores of a granule to allow for the dissolution of that particle

Pet. 4–5 (citing Ex. 1003 ¶¶ 53–55).

Patent Owner “agrees with the constructions adopted by the district court” and applies the additional constructions proposed by Petitioner for “wetting agent,” “binding agent,” and “disintegrating agent” as set forth above. Resp. 12. Patent Owner also agrees that the preamble of the challenged claims is not limiting. *See* Resp. 12; Tr. 50:17–20, 51:3–5 (Patent Owner’s counsel stating “[w]e agree that the preamble is not limiting. . . .”). Patent Owner does not propose any additional terms for construction.

The parties have not identified any disputes regarding the construction of any term in the challenged claims. As noted in the Institution Decision, the constructions proposed in the Petition “appear to be consistent with the cited portions of the intrinsic record of the ’685 patent as well as the usage of similar terms in the asserted prior art.” Dec. 29. Neither party disputes this, nor suggests that any of these terms have a different meaning in the art. Accordingly, we adopt the claim constructions noted above for our analysis.

We also adopt the parties’ position that the preamble of claims 1 and 7 is not limiting beyond the term “granule.” Claims 1 and 7 each recite “[a] chemically stable dry flow, low compact, dust free soluble phosphoramidothioate granule consisting of” the set of ingredients recited in body of those claims. Ex. 1001, 7:44–45, 8:19–20. While the wherein clause describing the dimensions of “said granule” relies on “granule” in the

preamble for antecedent basis, none of the other preamble language appears to be necessary to breathe life and meaning into the claim, and the claim body recites a structurally complete invention. *See Catalina Marketing Int'l, Inc. v. Coolsavings.com, Inc.*, 289 F.3d 801, 808 (Fed. Cir. 2002) (explaining that a preamble is limiting if it is “necessary to give live, meaning, and vitality to the claim,” but “is not limiting where a patentee defines a structurally complete invention in the claim body”) (internal quotations omitted). Thus, we agree with the parties, and the district court in the related litigation, that the preamble language other than “granule” is not limiting.

Moreover, we agree that the transitional phrase “consisting of” in claims 1 and 7 creates a “very strong presumption that the claim element is ‘closed’ and therefore exclude[s] any elements, steps, or ingredients not specified in the claim.” Pet. 5; Resp. 12 (both quoting *Multilayer Stretch Cling Film Holdings, Inc. v. Berry Plastics Corp.*, 831 F.3d 1350, 1358 (Fed. Cir. 2016)). Neither party seeks to overcome that presumption, nor do we see any basis for doing so based on the record before us. Accordingly, we construe the challenged claims to be limited to granule compositions containing only those ingredients recited in the claims.

There are no other terms that need to be construed to resolve the issues presented in this proceeding. Therefore, we do not construe any additional terms. *See, e.g., Nidec Motor Corp. Zhongshan Broad Ocean Motor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (“[W]e need only construe terms ‘that are in controversy, and only to the extent necessary to resolve the controversy.’” (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999))).

D. References Relied Upon

i. Misselbrook

Misselbrook is a patent issued on May 14, 2002, from a series of applications dating back to November 6, 1996. Ex. 1005, 1. Petitioner asserts that Misselbrook qualifies as prior art under 35 U.S.C. § 102(e). Pet. 2. Patent Owner does not dispute, and we agree, that Misselbrook is prior art to the '685 patent.

Misselbrook describes “soluble granule (SG) pesticidal formulation[s] comprising a water-soluble pesticide and a water-soluble filler.” Ex. 1005, Abstr. According to Misselbrook, these granule formulations provide “efficacy equal to the corresponding liquid formulation, yet with improved handler safety, such as lower eye irritation.” *Id.*; *see also id.* at 2:4–21 (describing “numerous advantages” its soluble granule formulation provides over liquid formulations).

While the examples in Misselbrook are directed to a different pesticide active ingredient (*see* Ex. 1005, 9:20–17:13), Misselbrook teaches that acephate is one of the “[e]xemplary water-soluble pesticides” for use in its granule compositions. *Id.* at 5:28–41. Misselbrook teaches “[t]he pesticidal compositions of the present invention comprise 0.1 to 90% by weight of a water-soluble pesticide . . . and 30 to 99.9% by weight of a water-soluble filler, preferably lactose (not to the exclusion of other ingredients).” *Id.* at 3:34–39; *see also id.* at 3:58–60 (“In one embodiment of the present invention the formulation contains about 0.1 to 90% by weight of the water-soluble pesticide.”).

Misselbrook also discloses “a more preferred embodiment” comprising:

0.1 to 60% by weight of a water-soluble pesticide . . . 40 to 99.9% by weight of a water-soluble filler; 0 to 50% by weight of a wetting surfactant; 0 to 50% by weight of a dispersing surfactant; 0 to 5% by weight of a defoaming agent (not to the exclusion of other ingredients).

Ex. 1005, 3:44–51. Misselbrook teaches that “[i]n addition to the pesticide, the water-soluble filler, the wetting surfactant, the dispersing surfactant and the defoaming agent, the instant pesticidal compositions may also appropriately contain stabilizers, synergists, coloring agents, etc.” *Id.* at 6:53–57.

Moreover, Misselbrook explains that “[t]he term ‘water-soluble filler’ as used herein includes any water soluble or water dispersible agent which may be employed to dilute the pesticide.” Ex. 1005, 6:1–3. Thus, in addition to sugars (e.g., “lactose, glucose, fructose, mannose”) Misselbrook teaches that other “[a]ppropriate water-soluble fillers” include “cellulose, calcium phosphates(s), inorganic water-soluble salts, and the like, and mixtures thereof.” *Id.* at 6:4–10.

ii. CN ’588

CN ’588 is a Chinese patent publication published on July 31, 1996. Ex. 1006, code (43). Petitioner asserts that CN ’588 is prior art under 35 U.S.C. § 102(b). Pet. 2. Patent Owner does not dispute, and we agree, that CN ’588 is prior art to the ’685 patent.

CN ’588 teaches that acephate “is less stable in pesticide preparations” than other “organophosphorous compounds” and “may be decomposed violently with the change of storage conditions, thereby unable

to effectively show activity all the time.” CN ’588, 4.⁷ According to CN ’588, “condensed sodium phosphate and/or synthetic silicic acid with alkylsilylated silanol groups on the surface can improve the stability of acephate in dry preparations.” *Id.*

To this end, CN ’588 describes a “dry pesticide preparation” wherein the “content of acephate” is

generally 0.5 to 99 (wt) %, preferably 5 to 95 (wt) %. The content of condensed sodium phosphate in the dry pesticide preparation is an amount that can stabilize acephate, generally 0.01 to 50 (wt) %, preferably 1 to 20 (wt) %, more preferably 1 to 10 (wt) %. The content of synthetic silicic acid with alkylsilylated silanol groups on the surface is an amount that can stabilize acephate, generally 1 to 99 (wt) %.

CN ’588, 5. In addition to these ingredients, CN ’588 teaches that its compositions may contain a wide variety of “solid carriers” as well as “adjuvants” such as “surface active agents, coating agents, fragrance and known stabilizers” in varying amounts. *Id.* CN ’588 further teaches that its dry pesticide preparation may be in the form of a “water soluble granule.” *Id.* at 6.

iii. JP ’902

JP ’902 is a Japanese patent publication published on December 9, 1997. Ex. 1008, code (43). Petitioner asserts that JP ’902 is prior art under 35 U.S.C. § 102(b). Pet. 2. Patent Owner does not dispute, and we agree, that JP ’902 is prior art to the ’685 patent.

⁷ As does the Petition, we cite to the page numbers of the translation in Exhibit 1007.

JP '902 describes “pesticide granules” that are “easy-to-use” because they “prevent caking during long-term storage and cause very little dusting during use.” JP '902, Abstr. JP '902 teaches that a number of pesticides, preferably acephate, can be formulated in such granules and that the “amount of pesticide active ingredient is about from 10 to 95% by weight relative to the entire formulation” (*Id.* ¶¶ 5–7) and “even more preferably about 50 to 95% by weight” (*Id.* ¶ 12).

In addition, JP '902 teaches that such granules may include other ingredients such as silica, surfactants, binders, colorants, stabilizers, and bulking agents. *Id.* ¶¶ 11–12. JP '902 discloses concentration ranges for some of these types of ingredients. *Id.* ¶ 12. For instance, JP '902 teaches the amount of surfactant “is usually about 10% by weight or less and preferably about 5% by weight or less,” the amount of binder “is usually about 20% by weight or less and preferably about 10% by weight or less,” and the amount of stabilizers “is about 10% by weight or less.” *Id.*

JP '902 provides a number of exemplary granule formulations containing acephate and other ingredients such as surfactants, binders, stabilizers, and bulking agents. *See, e.g.*, JP '902 ¶¶ 9, 11, 18, 24. The Petition cites two of those examples in particular, i.e., Reference Example 4 and Example 6, which are described as containing “95% Acephate.” Pet. 17–18 (citing Ex. 1008 ¶¶ 18, 24). Specifically, JP '902 describes these examples as follows:

[Reference Example 4]

Preparation of Granules D Containing 95% Acephate

After thoroughly mixing together 95 parts acephate, 0.5 parts Newpol PE-64 surfactant, 1.0 part dextrin, 0.1 part Cyanine

Green G, and 3.4 parts lactose, 8 parts water was added per 100 parts by weight formulation in terms of solid content, and the resulting mixture was prepared in the same manner as Reference Example 1 to obtain Granules D containing 95% acephate.

[Example 6]

Preparation of Granules J Containing 95% Acephate

After thoroughly mixing together 95 parts acephate, 0.2 parts Dixsol WK surfactant, 0.1 part Cyanine Green G, 0.5 part Aerosil 200, and 4.2 parts lactose, 4 parts water was added per 100 parts by weight formulation in terms of solid content, and the resulting mixture was prepared in the same manner as Reference Example 1 to obtain Granules J containing 95% acephate.

JP '902 ¶¶ 18, 24.

JP '902 describes disintegration tests conducted on these and other example granules and discloses the results of those tests in Table 1. JP '902 ¶¶ 28–32. JP '902 reports that “[i]t is clear from Table 1 that the granular water-soluble powders of the present invention have excellent disintegration in water, dispersibility in water, and caking properties.” *Id.* ¶ 31. In addition, JP '902 explains these “pesticide granules . . . prevented caking during long-term storage,” “had fewer harmful chemical effects,” and “readily disintegrated and dispersed in water to create easy-to-use formulations that can be used in place of conventional powdery water-soluble formulations and liquid formulations.” *Id.* ¶ 32.

iv. Mayer

Mayer is a patent issued on February 29, 2000. Ex. 1010, code (45). Petitioner asserts that Mayer qualifies as prior art under 35 U.S.C. § 102(b).

Pet. 2. Patent Owner does not dispute, and we agree, that Mayer is prior art to the '685 patent.

Mayer teaches that “[m]any solid crop protection agents are currently supplied in the form of wettable powders” containing an “insecticidal active ingredient[] . . . together with inert fillers, such as chalk or kaolin, in particular surface active substances, in the preparation of such formulations so that the formulations are well wetted and dispersed in water.” Ex. 1010, 1:9–15. According to Mayer, such formulations “tend to foam” and “develop dust when added to water in the spray tank.” *Id.* at 1:15–30. Thus, Mayer proposes to teach “solid formulations for crop protection with reduced tendency of foaming and dustiness.” *Id.* at 1:31–34, 14:49–61 (claim 1 reciting “[a] low dustiness water-dispersible granule formulation”).

Mayer teaches that the amount of “active ingredient” in its granule formulations “is, as a rule, 30 to 90% by weight, preferably 50 to 87% by weight, and, in particular, 70 to 85% by weight of total composition.” Ex. 1010, 4:4–6. In addition, Mayer teaches that its formulations contain a “defoaming agent and/or foam breaking agent” in an amount of “0.05 to 10% by weight.” *Id.* at 4:26–29. Mayer further teaches these formulations also include “formulation auxiliaries selected from the group consisting of dispersants, agglomeration auxiliaries, stabilizers, wetting agents, disintegrants and fillers” in varying amounts. *See id.* at 4:43–58.

Mayer also describes the preparation of “cylindrical granules.” *See* Ex. 1010, 5:65–6:23; 9:28–49 (Ex. 1). Mayer describes extruding the mixed ingredients “through a die or screen” with “die holes rang[ing] in diameter from 0.25 to 7 mm, and preferably, from 0.4 mm to 2 mm” and, after drying, sieving the product “on 0.4 and 2.0 mm sieves” to obtain granules. *Id.*

E. Ground 1: Obviousness over Misselbrook, CN '588, and JP '902

Petitioner contends that claims 1–4 and 7–12 are unpatentable as obvious over Misselbrook, CN '588, and JP '902. *See* Pet. 14–39. Patent Owner disputes this. *See* Resp. 13–37. As explained below, Petitioner has shown by a preponderance of the evidence that claims 1–4 are unpatentable, but has not shown by a preponderance of the evidence that claims 7–12 would have been obvious over these references.

i. Claim 1

Claim 1 recites a granule “consisting of” acephate and five other ingredients—a dispersing agent, a wetting agent, an antifoaming agent, a stabilizer, and fillers. Ex. 1001, 7:44–8:4. Petitioner must show that a granule containing only those ingredients, in the recited amounts, and in the recited dimensions would have been obvious.

To do so, Petitioner relies primarily on Misselbrook’s teaching of a granule containing a pesticide such as acephate and four other ingredients: a wetting surfactant, a dispersing surfactant, a defoaming agent, and a water-soluble filler. *See* Pet. 19–33. Petitioner contends, supported by the testimony of Mr. Geigle, that these ingredients respectively correspond to the “wetting agent,” “dispersing agent,” “defoaming agent” and “fillers” recited in claim 1. *Id.* at 22–27, 29–30; Ex. 1003 ¶¶ 83, 86, 89, 95. Petitioner also points out that Misselbrook teaches ranges for the amount of pesticide and these other ingredients that overlap with the corresponding weight percentage ranges in claim 1. *See* Pet. 23, 25, 26. Petitioner contends that the overlapping ranges in Misselbrook and the other references present a prima facie case of obviousness for the corresponding ranges in

claim 1. *See id.* at 13 (citing *In re Peterson*, 315 F.3d 1325, 1330 (Fed. Cir. 2003); *In re Wertheim*, 541 F.2d 257, 267 (CCPA 1976)).

For the “stabilizer” limitation, Petitioner directs us to Misselbrook’s teaching that, in addition to a granule containing the ingredients noted above, the “instant pesticidal compositions may also appropriately contain stabilizers, synergists, coloring agents, etc.” Pet. 27–28 (quoting 1005, 6:53–57). Petitioner further relies on teachings in CN ’588 and JP ’902 regarding the use of stabilizers in acephate compositions. *Id.* (citing CN ’588, 5; JP ’902 ¶¶ 11, 12, 20). In particular, Petitioner cites CN ’588’s teaching that “acephate is less stable in pesticide compositions” and “may decompose violently” to show that a POSITA would have been motivated to include a stabilizer in Misselbrook’s granule. *Id.* at 16 (quoting CN ’588, 4–5). According to Petitioner, a POSITA would have been motivated to include a stabilizer because they “would know that acephate may decomposed under changed storage conditions as compared to other organophosphorus compounds, and that adding a stabilizer could avoid such decomposition.” *Id.* at 28–29 (citing CN ’588, 4; Ex. 1003 ¶ 94).

For the “85-98% w/w . . . Acephate” limitation (Ex. 1001, 7:46–64), Petitioner argues that all three of the cited references disclose pesticide ranges that overlap with the recited acephate range, creating a presumption that range would have been obvious. *See* Pet. 14, 19–21 (citing Ex. 1005, 3:34–39; CN ’588, 5; JP ’902 ¶¶ 7, 12, 24). Petitioner also relies on working examples in JP ’902 that describe granules containing 95% acephate and some of the ingredients recited in claim 1. *Id.* at 20 (citing JP ’902 ¶¶ 18, 24). Petitioner contends that “[t]o allay the costs of granulation, a POSITA would aim for higher acephate concentrations to reduce manufacturing

costs” and therefore would have been motivated to formulate Misselbrook’s granule with an amount of acephate in the claimed range. *Id.* at 21 (citing Ex. 1005, 2:14–18; Ex. 1003 ¶ 74). According to Petitioner,

[a] POSITA would be motivated to use routine experimentation to determine a formulation allowing for high weight percentages of acephate to achieve cost savings. A POSITA would also possess a reasonable expectation of success in achieving higher acephate concentrations and would not view the concentration as unexpected or producing unexpected results based on at least JP ’902’s 95% acephate granules.

Id. at 21–22 (citing Ex. 1003 ¶¶ 80–82).

Regarding the concentration ranges for the other ingredients, Petitioner again relies on overlapping ranges disclosed in the cited references. *See* Pet. 23–30. Moreover, Petitioner alleges it would have been obvious to optimize the amounts of these ingredients to the recited ranges in order to accommodate a higher percentage of acephate. For example, regarding the amount of dispersing agent, Petitioner contends:

a POSITA would be motivated to achieve higher acephate concentrations, thereby decreasing other ingredients’ concentrations. A POSITA would know dispersing agents affect the performance of soluble granules when mixed in water for application. Because of the similarities among Misselbrook, CN ’588, and JP ’902 . . . a POSITA would use their disclosed ranges as a starting point for running routine tests and experiments to optimize the workable range of dispersing agent—including suspensibility tests (to determine the extent particles remain suspended when a granule is in solution) and wet sieving tests (to determine the amount of non-dispersible material in the granule). A POSITA would use these results to optimize a range for the dispersing agent, would possess a reasonable expectation of success in doing so, and would not view achieving the claimed range as unexpected or producing

unexpected results, since soluble granules generally used 1-6% by weight dispersing agent at that time.

Id. at 23–24 (citing Ex. 1003 ¶ 85, Ex. 1026, Ex. 1023, Ex. 1024, 28).

Petitioner offers similar arguments and evidence for the weight percentage ranges for the wetting agent (Ex. 1003 ¶ 88), antifoaming agent (*id.* ¶ 90), and stabilizer (*id.* ¶ 94), and fillers (*id.* ¶ 95).

Finally, Petitioner offers evidence that granules having the dimensions recited in claim 1, i.e., “a length of 1.5-3.0 mm and a diameter of 0.5-1.5 mm,” would have been obvious over these references. Mr. Geigle, testifies that a POSITA would have understood that granules produced by the processes described in Misselbrook and JP ’902 would have had a “0.8 mm diameter . . . with lengths between 0.5 to 5.0 mm.” Ex. 1003 ¶¶ 96–97, 99. Mr. Geigle further testifies that the recited dimensions “are not unexpected” and represent “typical diameter lengths used by manufacturers of water dispersible or soluble granules at the time.” *Id.* ¶ 100.⁸

Having considered the entire record, we find Petitioner’s contentions above to be credible and sufficiently supported by the evidence. As an initial matter, we agree that the combined references teach or suggest an acephate granule containing only the ingredients in claim 1. Misselbrook discloses a granule containing acephate and four of the five ingredients (i.e., a dispersing agent, a wetting agent, an antifoaming agent, and fillers) recited

⁸ Patent Owner does not dispute Mr. Geigle’s testimony that a POSITA would understand that the granulation processes described in Misselbrook and JP ’902 would produce granules having the recited dimensions and that such dimensions were typical for pesticide granules in the prior art. *See generally* Resp. We find Mr. Geigle’s testimony on this point credible and sufficient to meet Petitioner’s burden for this limitation.

in claim 1 and identifies the fifth ingredient, a stabilizer, as a type of ingredient its granules “may also appropriately contain.” Ex. 1005, 3:44–51, 6:53–57. Petitioner persuasively shows that a POSITA would have been motivated to add a stabilizer because acephate was known to have stability issues and both CN ’588 and JP ’902 teach the use of a stabilizer in acephate compositions, including granules. CN ’588, 4–5; JP ’902 ¶ 12. This combination results in a granule consisting of acephate and only the five other ingredients recited in claim 1.

Petitioner also demonstrates that the combined references teach or suggest the concentration ranges in claim 1. Beginning with the acephate range, Misselbrook teaches compositions containing between “0.1 to 90% by weight” pesticide. Ex. 1003, 3:34–39, 3:58–60. CN ’588 and JP ’902 disclose similar acephate ranges. *See* CN ’588, 2 (acephate preferably 5 to 95% wt); JP ’902 ¶ 12 (acephate preferably 50 to 95% wt). The upper end of the prior art ranges overlap with the 85–98% range in claim 1, creating a presumption that the claimed range would have been obvious. *See E.I. DuPont de Nemours & Co. v. Synvina C.V.*, 904 F.3d 996, 1006 (Fed. Cir. 2018) (explaining that “a prima facie case of obviousness typically exists when the ranges of a claimed composition overlap the ranges disclosed in the prior art” and “such overlap creates a presumption of obviousness”) (quotations omitted). Patent Owner notes that a preferred embodiment in Misselbrook on which some of Petitioner’s contentions rely discloses a lower pesticide range that does not overlap with the claimed range. *See* Resp. 7–8. But Misselbrook’s teachings are not limited to that preferred embodiment. And “the fact that a specific embodiment is taught to be preferred is not controlling, since all disclosures of the prior art, including

unpreferred embodiments, must be considered.” *Merck & Co., Inc. v. Biocraft Labs., Inc.*, 874 F.2d 804, 807 (Fed. Cir. 1989) (quoting *In re Lamberti*, 545 F.2d 747, 750 (CCPA 1976)). Thus, Patent Owner’s focus on Misselbrook’s preferred embodiment does not negate the presumption arising from the broader, overlapping range taught elsewhere in Misselbrook and in the other cited references.

In addition to the presumption, Petitioner persuasively shows that it would have been obvious to increase the acephate concentration in Misselbrook’s granules to the recited range. Mr. Geigle notes the higher costs associated with preparing granule versus liquid and powder formulations and explains that a POSITA would have sought to offset those costs by increasing the acephate concentration in the granules so as to achieve the benefits of a granule formulation at a lower cost. Ex. 1003 ¶¶ 25, 27–28, 59–60. He further explains that this would have motivated a POSITA to maximize the acephate concentration in Misselbrook’s granules and that there would have been a reasonable expectation of successfully achieving a high acephate content within the claimed range given that JP ’902 disclosed working examples of 95% acephate granules. *Id.* ¶¶ 74, 80–82. We find this evidence credible and sufficient to support Petitioner’s contentions regarding the “85-98% w/w” acephate limitation.

Petitioner also meets its burden for the other ranges recited in claim 1. It is well-established that “where the general conditions of a claim are disclosed in the prior art, it is not inventive to discover the optimum or workable ranges by routine experimentation.” *In re Aller*, 220 F.2d 454, 456 (CCPA 1955). Here, Petitioner identifies where the cited art discloses overlapping ranges for each of other ingredients in claim 1, thereby

demonstrating that the “general conditions” regarding the use of these ingredients in such compositions was known. *In re Applied Materials, Inc.*, 692 F.3d 1289, 1295 (Fed. Cir. 2012) (noting that any overlap between the claimed and disclosed ranges serves to disclose the “general conditions” of the claim). “Such overlap itself provides sufficient motivation to optimize the ranges.” *Id.* Moreover, we credit Mr. Geigle’s testimony that a POSITA would have understood because the concentration of acephate was increased the concentration of the other ingredients would necessarily decrease and that a POSITA would have been able optimize these amounts using standard techniques and testing. Ex. 1003 ¶¶ 85–113. Thus, Petitioner has demonstrated that these ingredient concentrations are result-effective variables that would have been obvious to optimize through routine experimentation.

Patent Owner does not seek to rebut Petitioner’s showing with evidence of unexpected results or other evidence of criticality relating to the ranges in claim 1. *See* Reply 8 (“[Patent Owner] does not, and has never, alleged that its invention claims an unexpected result.”); Sur-Reply 4 (responding by challenging Petitioner’s “*prima facie* case” and agreeing “there is no need for the Board to look to unexpected results or other indicia of nonobviousness”). Rather, Patent Owner attacks the sufficiency of Petitioner’s showing in several respects. We address Patent Owner’s arguments below to explain why they are unavailing.

Patent Owner argues that the prior art evidences “difficulties with preparing granules containing 85-98% acephate.” *See* Resp. 3–7 (citing Ex. 2003, Ex. 2005 and Ex. 2006). According to Patent Owner, the Petition cites only “a *single* prior art reference describing granules containing 85-

95% acephate, JP '902.” *Id.* at 7. Patent Owner argues that Petitioner’s other references “do not provide any working examples of a granule containing acephate much less 85-98% acephate” and do not “propose any process, method, or formulation to overcome the difficulties” with formulating acephate granules. *Id.*

Regarding the two working examples of 95% acephate granules in JP '902, Patent Owner argues that one does “not provide a workable option for granules containing 85-98% acephate” because “most of the product remained caked at the bottom of the container.” Resp. 8 (citing JP '902 ¶¶ 15, 24). Patent Owner acknowledges that the other example (Example 6) is reported to have “acceptable disintegration properties,” but argues that because JP '902 “fails to provide any information regarding its chemical stability” it is “unclear whether JP '902 provides a workable solution.” *Id.* at 9. Patent Owner further contends that Petitioner “does not rely on Example 6 as the starting point in its arguments, and . . . does not attempt to show any path (much less a scientifically valid path) for modifying Example 6 to arrive at the granules claimed in the '685 patent.” *Id.* at 9. From this Patent Owner concludes “the prior art highlights the unpredictability and difficulties with preparing granules containing 85-98% acephate and establishes that selecting and optimizing ingredients and amounts to prepare such granules was not a simple matter of routine experimentation.” *Id.*

Patent Owner’s arguments are unpersuasive for several reasons. First, as Patent Owner acknowledges, JP '902 discloses actual working examples of acephate granules, including Example 6 that contains 95% acephate in combination with some of the other ingredients taught in Misselbrook and

recited in claim 1. JP '902 ¶¶ 18, 24; Ex. 1003 ¶¶ 83, 86, 95 (explaining these granules also contain a dispersing agent, wetting agent, and fillers). Nothing in JP '902 suggests these exemplary granules were difficult to prepare or otherwise unsuccessful. To the contrary, JP '902 reports that the granule in Example 6 had “excellent disintegration in water, dispersibility in water, and caking properties” and provides an “easy-to-use formulation[] that can be used in place of conventional powdery water-soluble formulations and liquid formulations.” JP '902 ¶¶ 31–32. Patent Owner’s argument that Petitioner does not use Example 6 as the “starting point” for its analysis misses the point. *See* Resp. 9. Petitioner relies on the 95% acephate granule in Example 6 to demonstrate both a motivation to increase the acephate content in Misselbrook’s granule and to show that a POSITA would have had a reasonable expectation of successfully achieving a high acephate concentration within the claimed range. *See* Pet. 21–22. We agree with Petitioner on both counts.

Second, Patent Owner’s focus on what it characterizes as a lack of relevant working examples in Misselbrook and CN '588 ignores the other teachings in these references. Both references teach acephate compositions in the form of granules. Ex. 1005, 2:55–59; 5:35–41; CN '588, 4–5. And both references teach that the amount of acephate in those granules may be as much as 90% by weight. Ex. 1005, 3:34–35, 58–60; CN '588, 4. The fact that, unlike JP '902, Misselbrook and CN '588 do not further exemplify these teachings in a working example does not negate the relevance of their teachings in an obviousness analysis. *See, e.g., Merck*, 874 F.2d at 807 (“all disclosures of the prior art . . . must be considered”).

Third, we credit Mr. Geigle’s testimony that pesticide granules and the general principles for making them were known in the prior art. *See* Ex. 1003 ¶¶ 24–29. This testimony is supported by the background teachings in Misselbrook and JP ’902. *See, e.g.*, Ex. 1005, 1:40–50 (referring to “methods for preparing water-soluble granules of pesticides [that] are known in the art”); JP ’902 ¶¶ 13–14 (describing methods “commonly used to prepare pesticide granules”).

Moreover, Petitioner presents evidence that there were already a number of EPA registrations for high acephate content pesticide compositions prior to 2001, including at least one that Patent Owner acknowledges is for a “granular product containing 85-98% acephate.” Ex. 1003 ¶ 23 (citing Exs. 1012–1014); Sur-Reply 5 (citing Ex. 2007 ¶¶ 66–71, Attachments A and B). These registrations are not themselves references for any of the grounds in the Petition, but their existence corroborates Mr. Geigle’s testimony that the general conditions of formulating acephate compositions were already known in the art. Such evidence also undermines Patent Owner’s argument that there were intractable difficulties to the formulation of acephate granules containing 85-98% acephate. For all these reasons, we credit Petitioner’s showing over Patent Owner’s evidence of the alleged difficulties in formulating such granules.

Patent Owner next argues that Petitioner’s combination of references would have led a POSITA to include a binding agent, which is not one of the ingredients recited in claim 1. Resp. 13–19. According to Patent Owner, “[b]inding agents were understood to reduce granule dustiness, which was and is an important safety consideration.” *Id.* at 14 (citing Ex. 2007 ¶¶ 83–88; Ex. 1024, 62, 64, 66–67). Patent Owner argues that the cited references

“teach[] away from preparing a granule without a binder” because “Misselbrook discloses granules containing water-soluble binders, preferably ‘lactose, sucrose, glucose, and the like.’” *Id.* at 15 (quoting Ex. 1005, 3:2). Patent Owner acknowledges that Misselbrook refers to these agents as ‘water-soluble fillers,’” but urges that a POS[IT]A “would have known that these agents assist in holding the particles together, i.e., they are binding agents.” *Id.* (citing Ex. 2007 ¶ 92). According to Patent Owner, “water-soluble binders are a key feature of the granules” taught in Misselbrook and “neither CN ’588 nor JP ’902 provide any scientific reason to *remove* Misselbrook’s water-soluble binding agent.” *Id.* at 17–18.

Petitioner does not dispute that lactose, sucrose, and glucose were known to be binding agents, but argues that Misselbrook also teaches the use of other water-soluble fillers that do not have binding properties. Reply 10–12. Specifically, Petitioner contends that “Misselbrook’s formulations teach using a ‘water soluble filler’ which ‘includes *any* water soluble or water dispersible agent which may be employed to dilute the pesticide.’” *Id.* at 10 (quoting Ex. 1005, 6:1–3 (emphasis added by Petitioner)). According to Petitioner,

A filler is a common, well-understood concept to a POSITA, and specific examples of non-binding-agent-fillers are not needed in order to disclose the concept of a filler. A POSITA would know fillers normalize the concentration of the active ingredient in a formulation, would reasonably expect success in doing so, and would not view this as unexpected or producing unexpected results.

Id. at 11 (citing Ex. 1003 ¶ 95). Petitioner further points to the disclosure of fillers “which are not also binding agents,” such as “inorganic salts” in CN ’588 and JP ’902. *Id.* (citing CN ’588, 2; JP ’902 ¶¶ 5, 11).

We agree with Petitioner that Misselbrook’s teachings are not limited to the types of water-soluble fillers that Patent Owner says are binding agents. In addition to lactose and other sugars, Misselbrook teaches that “microcrystalline cellulose, powdered cellulose, calcium phosphate(s), [and] inorganic water-soluble salts” are also “[a]ppropriate water-soluble fillers” for its granules. Ex. 1003, 6:4–10. Nothing in the record suggests that such fillers are binding agents. Moreover, Misselbrook’s express teaching that “*any* water soluble or water dispersible” diluent can be used as fillers (Ex. 1003, 6:1–3 (emphasis added)) rebuts Patent Owner’s argument that a POSITA would understand fillers with binding properties like lactose, sucrose, and glucose to be a “key feature” of those granules. Resp. 18. Thus, the combination of references in Petitioner’s ground would not, as Patent Owner contends, have led a POSITA to include a binding agent.

Furthermore, we are not convinced that the premise for Patent Owner’s argument—that the “consisting of” language in claim 1 excludes Misselbrook’s use of lactose as a filler—is correct. “Consisting of” excludes unrecited ingredients. But “fillers to make 100%” are one of the ingredients recited in claim 1. Ex. 1001, 8:2. Mr. Geigle persuasively explains that the fillers taught in Misselbrook, including lactose, are used “to normalize the concentration of the active ingredient” by “adding an amount . . . to make 100%.” Ex. 1003 ¶ 95. This would seem to meet the plain meaning of “fillers to make 100%,” as recited in claim 1.⁹ Thus, we do not understand “consisting of” to exclude such fillers from the granule of claim 1 simply

⁹ Petitioner does not propose a specific construction for this claim term, but states it should be given its plain meaning. Pet. 3. Patent Owner does not respond to this contention, nor propose its own construction.

because a POSITA would understand that the same ingredient may also serve a different purpose. That said, because Misselbrook's teachings are not limited to lactose, and Petitioner has sufficiently shown that it would have been obvious to use both non-binding and binding fillers in the asserted combination, Patent Owner's argument would remain unavailing even if claim 1 were construed as Patent Owner suggests.

Patent Owner further argues that there is "no basis to expect that a POS[IT]A would 'optimize' the level of stabilizer to 0.01-1%" because the cited art either "does not suggest any amount" of stabilizer to use or it suggests the use of a higher concentration. *See* Resp. at 19–22. According to Patent Owner:

CN '588 reports that acephate decomposed, even when combined with stabilizers at 20-30%, well above the claimed range of 0.01-1%. Thus, while CN '588 generally states that stabilizers could be used at "0.01 to 10 parts" by weight relative to acephate, CN '588 provides no motivation for selecting a stabilizer at or near the low end of this range. Thus, CN '588 undermines [Petitioner's] notion that a POS[IT]A would somehow "optimize" the level of stabilizer to 0.01-1% based on accelerated aging tests.

Id. at 20 (citing Ex. 2007 ¶¶ 119–22).

We disagree. As explained above, both CN '588 and JP '902 teach stabilizer ranges that encompass or overlap the recited range at their lower end. *See, e.g.*, CN '588, 5 (a stabilizer in "an amount that can stabilize acephate, generally 0.01 to 50 (wt) %"); JP '902 ¶ 12 ("The amount of other additive such as stabilizers and preservatives relative to the overall formulation is about 10% by weight or less."). CN '588 teaches that an amount of stabilizer in this range "can stabilize acephate" in a dry pesticide preparation such as a granule. CN '588, 5. Moreover, we credit Mr.

Geigle’s testimony that a POSITA would understand that as the concentration of acephate was increased, the amount of the other ingredients, including the stabilizer, would have to decrease. *See, e.g.*, Ex. 1003 ¶ 94. Such evidence demonstrates a motivation to optimize the concentration of stabilizer toward the lower end of the ranges taught in these references.

The results of CN ’588’s accelerated aging tests, which Patent Owner refers to in its Response, do not overcome Petitioner’s showing. As an initial matter, the data Patent Owner relies upon shows that the compositions containing a stabilizer had a lower decomposition rate than the control samples. CN ’588, 7–8; Ex. 2007 ¶ 120 (highlighting data). Thus, the data do not undermine CN ’588’s teaching of effective stabilizer ranges, but rather support the reference’s thesis that adding such stabilizers “can improve the stability of acephate in dry preparations.” CN ’588, 4. Moreover, as Petitioner points out, claim 1 does “not claim a *stable* acephate granule” that undergoes no decomposition whatsoever in accelerated aging tests, but rather a “granule that *includes a stabilizer*.”¹⁰ Reply 19. And Petitioner has sufficiently shown that a POSITA would have been motivated to add a stabilizer, been able to optimize the amount using routine techniques, and had a reasonable expectation of achieving some improvement by doing so even at the lower end of the prior art ranges.

Finally, Patent Owner’s arguments that Petitioner’s ground is based on

¹⁰ While the claim preamble refers to a “stable” granule, Patent Owner agrees this language is not limiting. Tr. 50:17–20, 51:3–5. Thus, Patent Owner has waived any argument that the challenged claims can be distinguished from the cited prior art based on language in the preamble.

improper hindsight are also unavailing. *See* Resp. 23–27. Patent Owner contends that a POSITA would not have looked to Misselbrook as the “starting point” for developing acephate granules because “Misselbrook’s disclosure focuses on emamectin benzoate, which is Misselbrook’s preferred, more preferred, even more preferred, and especially preferred pesticide.” *Id.* at 23. But while the Misselbrook inventors may have preferred emamectin benzoate, they also identify “acephate” as an “[e]xemplary water-soluble pesticides which may be employed in the present invention.” Ex. 1005, 5:35–41. It is well settled that “the fact that a specific embodiment is taught to be preferred is not controlling, since all disclosures of the prior art, including unpreferred embodiments, must be considered.” *Merck*, 874 F.2d at 807 (Fed. Cir. 1989) (quotations omitted). Likewise, even if a reference’s “principal focus” lies elsewhere, it does not “negate” additional teachings that other embodiments are also effective. *See In re Inland Steel Co.*, 265 F.3d 1354, 1361 (Fed. Cir. 2001). This maxim that all teachings in a reference must be considered is all the more true here because Misselbrook identifies acephate as one of a handful of exemplary pesticides for its granule formulations—not an unpreferred option. Thus, contrary to Patent Owner’s argument, Petitioner has shown that Misselbrook is a valid starting point for its obviousness analysis.

Patent Owner further contends the ground is hindsight-driven because the references “do not teach any granule containing the five excipients—and only the five excipients—recited in claim 1 of the ’685 patent.” Resp. 25. To this end, Patent Owner identifies “numerous specific choices” it contends are not taught in the references and that a POSITA would have to make to arrive at the granule of claim 1. *Id.* at 25–27.

We do not agree with Patent Owner. As explained above, Petitioner demonstrates that Misselbrook teaches an acephate granule containing four of the excipients in claim 1 and sets forth sufficient reasoning with rational underpinning to include the fifth as taught by both CN '588 and JP '902. Petitioner also persuasively shows that it would have been obvious to optimize the concentrations within the recited ranges. The flaw in Patent Owner's argument is that, to the extent they are relevant, the choices it contends one would have had to make to arrive at the claimed granule are all taught or suggested by the combination of references, and in most instances by Misselbrook itself. Moreover, Petitioner has shown that any leaps a POSITA would have to make are not nearly as great as Patent Owner suggests and well within the knowledge of one of ordinary skill using routine experimentation. To the extent there are any disagreements between the parties' experts on this point, we credit the testimony of Mr. Geigle and the evidence and rationale offered in the Petition.

In sum, Petitioner addresses every element of claim 1 and persuasively demonstrates that a POSITA would have been motivated to combine the cited teachings with a reasonable expectation of success. After considering the full trial record, we determine Petitioner has shown by a preponderance of the evidence that claim 1 is unpatentable as obvious over Misselbrook, CN '588, and JP '902.

ii. Claims 2–4

Claims 2–4 depend from claim 1. These claims further limit the “dispersing agent” (claim 2), “wetting agent” (claim 3), and “antifoaming

agent” (claim 4) in claim 1 to a more particular category of such ingredients. *See* Ex. 1001, 8:5-14.

Petitioner identifies where examples of ingredients within each of the categories recited in claims 2–4 are taught in Misselbrook and, in some instances, the other cited references as well. *See* Pet. 37 (citing Ex. 1025, 2–3 (sodium lauryl sulfate); Ex. 1005, 6:36–37; Ex. CN ’588, 5, JP ’902 ¶ 10 for claim 2); 37–38 (citing Ex. 1005, 4:1–9 (sodium alkyl naphthalene sulfonate); CN ’588, 5; Ex. 1009 ¶ 10 for claim 3); 38 (citing Ex. 1005, 3:13–15 (methylated silicones)). Petitioner also offers testimony from Mr. Geigle supporting these arguments. Ex. 1003 ¶¶ 102–103 (claim 2); 104–105 (claim 3); 106–107 (claim 4). Patent Owner does not dispute Petitioner’s contentions regarding the additional limitations recited in claims 2–4. *See generally* Resp.

Petitioner’s contentions are sufficiently supported by the record and persuasive. Thus, we find that the additional limitations recited in claims 2–4 are taught in the cited references, and it would be obvious to use such ingredients in the asserted combination. To the extent Patent Owner asserts the same arguments for claims 2–4 that it does for claim 1, those arguments are unavailing for the reasons noted above. Accordingly, Petitioner has shown by a preponderance of the evidence that claims 2–4 are unpatentable as obvious over Misselbrook, CN ’588, and JP ’902.

iii. Claim 7

We reach a different conclusion on independent claim 7. Like claim 1, claim 7 is a “consisting of” claim. Claim 7 recites all of the ingredients and ranges in claim 1 and further recites a specified amounts of a

“binding agent” and a “disintegrating agent.” Ex. 1001, 8:41–43. For the ingredients recited in both claims, Petitioner relies on the same evidence and arguments discussed above for claim 1. *See* Pet. 14–37. Petitioner presents additional arguments regarding the binding agent and disintegrating agent limitations. *Id.*

We have considered these arguments and determine that Petitioner does not sufficiently show that a POSITA would have been motivated to combine a disintegrating agent with the other ingredients in claim 7. Because this issue is dispositive of Petitioner’s challenge to claim 7, we focus on the parties’ arguments regarding the disintegrating agent limitation in our analysis below.

Petitioner acknowledges that neither Misselbrook, nor JP ’902, teaches a disintegrating agent as an ingredient in their acephate compositions. Pet. 35–36; Tr. 25:12–26:1 (Petitioner’s counsel agreeing that Misselbrook does not identify disintegrating agents). According to Petitioner, “CN ’588 teaches adding ‘bentonite’ as a ‘solid carrier’ to dry acephate formulations” and a POSITA would know bentonite “can be used as a disintegrating agent.” Pet. 35–36. Petitioner argues that a POSITA would have been motivated to add a disintegrating agent, such as bentonite, to Misselbrook’s granule because he or she “would know the need for disintegrating agents varies based on a granules compactness and ingredients, and that adding a binding agent may require adding a disintegrating agent to assist in granule dissolution.” *Id.* at 36 (citing Ex. 1003 ¶ 112). Petitioner further contends “a POSITA would be motivated to run routine tests and experiments to optimize the workable range of the disintegrating agent—including testing the rate of dispersion” as

in the disintegration tests described in JP '902. *Id.* (citing Ex. 1003 ¶ 113; JP '902 ¶ 28).

Patent Owner disagrees, urging that “[t]he art provides no scientific rationale for adding a disintegrating agent to Misselbrook’s granules.” Resp. 32. According to Patent Owner, both Misselbrook and JP '902 teach acephate granules without a disintegrating agent and, “[w]hile CN '588 mentions disintegrating agents among a laundry list of solid carriers,” it “does not disclose any benefit associated with adding a disintegrating agent to a granule of the type disclosed in Misselbrook.” *Id.* at 32–33.

Moreover, Patent Owner contends that a POSITA “would have understood that the granules in Misselbrook did not require a disintegrating agent based on at least JP '902.” *Id.* at 33 (citing Ex. 2007 ¶ 156). Patent Owner’s expert, Dr. Rockstraw, explains that the disintegration tests in JP '902 show that “granules containing water-soluble pesticides, water-soluble binders, and *no* disintegrating agent,” like those disclosed in JP '902 and Misselbrook, “had acceptable disintegrating properties when mixed with water, i.e., no disintegrating agent was required.” Ex. 2007 ¶ 155. As such Patent Owner argues that “[c]onsistent with [Petitioner’s] theme of efficiency and reducing costs, a POSA would not seek to add an extra ingredient such as a disintegrating agent when it was not required.” Resp. 33.

In its Reply, Petitioner responds largely by repeating its arguments from the Petition. *See* Reply 22–24. Petitioner does not directly respond to Patent Owner’s argument and Dr. Rockstraw’s testimony that the results of JP '902’s disintegration tests suggest that a disintegrating agent is not

required even when those granules contain an ingredient with binding properties such as lactose. *Id.*

Having considered the record as a whole, we conclude that Petitioner has not met its burden for this element. The record shows that disintegrating agents such as bentonite were known in the art. But obviousness requires more than showing that a claim element was known in the prior art, Petitioner must also demonstrate that a motivation for combining that element with the others recited in the claim. *KSR*, 550 U.S. at 418.

Here, Petitioner relies on the following testimony of Mr. Geigle to try to show the required motivation:

A POSITA in 2001 would know that the need for a disintegrating agent varies based on the compactness of a granule, and the other ingredients used in the formulation. A POSITA would know that adding a binding agent may require adding a disintegrating agent to assist in the dissolution of a granule in solution.

Ex. 1003 ¶ 112.

This testimony and the arguments Petitioner premises on it are not persuasive. As an initial matter, the testimony is equivocal. Mr. Geigle states only that a POSITA would know that a disintegrating agent “may” be required. Ex. 1003 ¶ 112. More importantly, Petitioner’s argument that a POSITA would have known that a disintegrating agent may be required does not account for disintegration tests reported in JP ’902. Those tests show that acephate granules containing several of the same ingredients recited in the challenged claims had “excellent disintegration in water, dispersibility in water, and caking properties” without the need for any disintegrating agent. JP ’902 ¶ 31; *see also id.* ¶ 32 (“[T]he pesticide granules readily disintegrated and dispersed in water”). This is true even though at least

some of those granules, including the 95% acephate granule in Example 6, also contain lactose—an ingredient both parties agree can act as a binding agent. *Id.* Petitioner does not explain why a POSITA would have understood that a disintegrating agent may be required in the presence of such a binding agent when there is data in the record suggesting that no disintegrating agent is required. For these reasons, we credit Dr. Rockstraw’s testimony that “a POS[IT]A would have expected that a disintegrating agent would not be required to achieve acceptable dissolution” and that “factors such as cost and efficiency would have discouraged adding extra, unnecessary ingredients.” Ex. 2007. ¶¶ 156, 248.

Moreover, Petitioner’s argument that it would have been obvious to add a disintegrating agent absent sufficient evidence of a need to do so is in tension with its other arguments. As Patent Owner points out, Petitioner advances a “theme of efficiency and reducing costs” both to show a motivation to increase the concentration of acephate and to optimize the concentrations of the other ingredients. *See* Resp. 33 (citing arguments in the Petition). That theme is at odds with Petitioner’s argument for adding a disintegrating agent because JP ’902 suggests such an agent is not necessary to produce a granule that readily dissolves. Petitioner has not sufficiently reconciled this tension, which further undermines its argument for adding a disintegrating agent to Misselbrook’s granule.

For these reasons, we find that Petitioner’s alleged motivation for including a disintegrating agent in its combination of references is not supported by a preponderance of the evidence. Accordingly, Petitioner has not shown sufficiently that claim 7 would have been obvious over Misselbrook, CN ’588, and JP ’902.

iv. Claims 8–12

Claims 8–12 depend from claim 7 and therefore also require a “disintegrating agent.” For the same reasons Petitioner has not carried its burden for claim 7, Petitioner has not shown that claims 8–12 are unpatentable over the cited references.

F. Petitioner’s Other Grounds

Petitioner’s other grounds are similar to its first ground and challenge the same claims, but add Mayer to the combination of Misselbrook and CN ’588 (ground 2) and Misselbrook and JP ’902 (ground 3). *See* Pet. 39–61. Having determined that Petitioner establishes by a preponderance of the evidence that the subject matter of claims 1–4 would have been obvious over Misselbrook, CN ’588, and JP ’902 we need not address Petitioner’s additional grounds challenging these claims. *See SAS Inst. Inc. v. Iancu*, 138 S. Ct. 1348, 1359 (2018) (holding that a petitioner “is entitled to a final written decision addressing all of the claims it has challenged”); *Boston Sci. Scimed, Inc. v. Cook Grp. Inc.*, 809 F. App’x 984, 990 (Fed. Cir. 2020) (nonprecedential) (“We agree that the Board need not address [alternative grounds] that are not necessary to the resolution of the proceeding.”).

As for claims 7–12, we again find that Petitioner’s alleged motivation for including a disintegrating agent is lacking. This is true for both ground two and ground three because Petitioner advances essentially the same arguments for this limitation in both grounds. *Compare* Pet. 50–51, *with id.* at 61–62. Our discussion and findings below apply to both grounds.

i. Claim 7

According to Petitioner,

Mayer teaches “[c]onventionally used disintegrants may be used in the solid formulation according to the present invention,” and that “[t]he solid formulation according to the present invention preferably comprises 1 to 10% by weight, in particular, 1 to 7.5% by weight of at least one disintegrant”—falling within the claimed range. Mayer describes examples of 75-85% and 70-85% active ingredient granules containing “1 to 5%” disintegrant and “1 to 7.5%” disintegrant, respectively.

Pet. 50; *see also id.* at 61 (referring to these teachings). Based on this “knowledge of disintegrating agents, the effect binders have on disintegration, the focus on higher acephate concentrations, and motivation to look to Mayer . . . to improve upon Misselbrook,” Petitioner argues that a POSITA would have been motivated to run routine tests to optimize the workable range of disintegrating agent starting at the range in Mayer. *Id.* at 50–51. Petitioner alleges the same motivation for adding a disintegrating agent to the combination in ground three, additionally pointing out that “JP ’902 describes testing 95% acephate granules for disintegration, consistent with routine tests a POSITA would know to conduct in 2001.” *Id.* at 61.

As it did in the first ground, Patent Owner responds that none of the cited references “provide[] any scientific rationale for adding a disintegrating agent to Misselbrook’s granules” and “[t]he mere fact” that the references “mention the possibility of adding a disintegrating agent to pesticide formulations does not establish that a POS[IT]A would have been motivated to add one to *Misselbrook’s* granules.” Resp. 49–50; 62–63. Patent Owner repeats its argument that “based on Misselbrook and JP ’902, a POS[IT]A would have understood that granules comprised largely of

water-soluble pesticides and water-soluble binding agents do not require disintegrating agents.” *Id.* at 50 (citing Ex. 2007 ¶¶ 154–56, 212).

According to Patent Owner, a POSITA “would have understood that Misselbrook’s granules did not require a disintegrating agent” and concerns regarding cost and efficiency would have discouraged the addition of an unnecessary ingredient. *Id.*; *see also id.* at 62–63 (same response for ground three).

Having considered the record as a whole, including the additional teachings cited in Mayer, we again determine that Petitioner has not carried its burden. It is true that Mayer teaches that a disintegrating agent is one of many types of ingredients that may be included in a pesticide granule. But, as Patent Owner points out, that teaching does not itself evidence a motivation to add a disintegrating agent to Misselbrook’s granule. *See* Resp. 49–50; 62–63.

The motivation Petitioner alleges for combining Mayer’s disintegrating agent in grounds two and three is the same one it offers for the first ground, i.e., that a POSITA would have known that a disintegrating agent may be required when a binding agent is used. As explained above, that argument is unpersuasive in view of the disintegration tests reported in JP ’902 and Dr. Rockstraw’s testimony that based on those tests a POSITA would expect acephate granules with excipients like those taught in Misselbrook and JP ’902 to dissolve acceptably without the need for a disintegrating agent. JP ’902 ¶¶ 31–32; Ex. 2007 ¶¶ 155–158, 209–212, 247. While Mayer describes pesticide granules and disintegrating agents generally, it does not mention acephate at all. *See* Pet. 41 (acknowledging that “Mayer does not explicitly name acephate”). Thus, Mayer does not

suggest a need for a disintegrating agent in an acephate granule, much less one with the excipients similar to those taught in Misselbrook and JP '902. Accordingly, we agree with Patent Owner that Mayer's teachings do not provide a persuasive rationale for adding a disintegrating agent to Misselbrook's granules and credit Dr. Rockstraw's testimony that "factors such as cost and efficiency would have discouraged adding extra, unnecessary ingredients" such as a disintegrating agent. Ex. 2007 ¶ 248.

For these reasons, as well as those explained in our analysis of claim 7 in the first ground, we find that the preponderance of the evidence does not support Petitioner's alleged motivation for combining a disintegrating agent with the other ingredients in its combination of references to show that claim 7 would have been obvious. Accordingly, Petitioner does not sufficiently show that claim 7 would have been obvious over either Misselbrook, Mayer, and CN '588 (ground two) or Misselbrook, JP '902, and Mayer (ground three).

ii. Claims 8–12

For the same reasons Petitioner has not carried its burden for claim 7, Petitioner has not shown by a preponderance of the evidence that claims 8–12 are unpatentable over the references in grounds two and three.

IV. CONCLUSION¹¹

Petitioner demonstrates, by a preponderance of the evidence, that claims 1–4 of the '685 patent are unpatentable, but does not establish the same for claims 7–12.

¹¹ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this

Claims	35 U.S.C. §	Reference(s)/ Basis	Claims Shown Unpatentable	Claims Not shown Unpatentable
1–4, 7–12	103(a)	Misselbrook, CN '588, JP '902	1–4	7–12
1–4, ¹² 7– 12	103(a)	Misselbrook, Mayer, CN '588		7–12
1–4, ¹³ 7– 12	103(a)	Misselbrook, JP '902, Mayer		7–12
Overall Outcome			1–4	7–12

V. ORDER

Accordingly, it is:

ORDERED that Petitioner has shown that claims 1–4 of U.S. Patent 7,473,685 B2 are unpatentable;

FURTHER ORDERED that Petitioner has not shown that claims 7–12 of U.S. Patent 7,473,685 B2 are unpatentable; and

Decision, we draw Patent Owner's attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. See 37 C.F.R. §§ 42.8(a)(3), 42.8(b)(2).

¹² As explained above, we do not reach this ground for claims 1–4.

¹³ As explained above, we do not reach this ground for claims 1–4.

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FURTHER ORDERED that, because this is a Final Written Decision, parties to this proceeding seeking judicial review of our Decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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